

Accelerating Linux Virtual Server with OpenNPU

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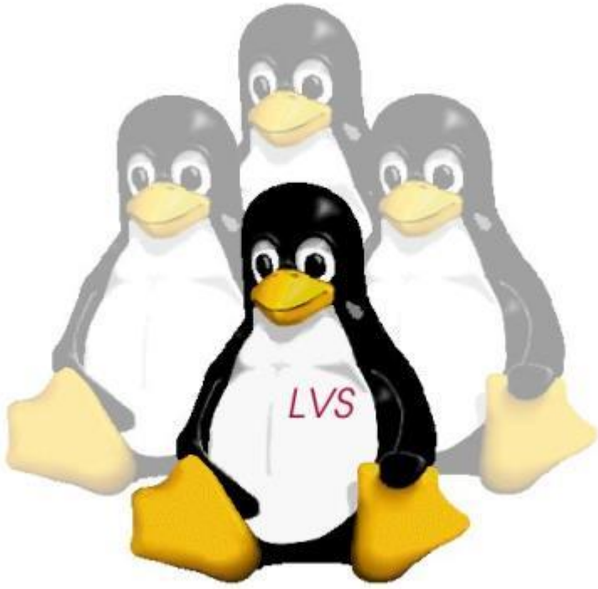
How it all started

辻斬り



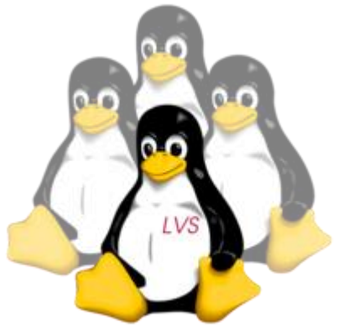
Tsujigiri (辻斬り or 辻斬 *tsuji-giri*, literally *crossroads killing*) is a Japanese term for a practice when a samurai, after receiving a new *katana* or developing a new fighting style or weapon, tests its effectiveness by attacking a human opponent, usually a random defenseless passer-by, in many cases during nighttime.

https://en.wikipedia.org/wiki/Tsujigiri



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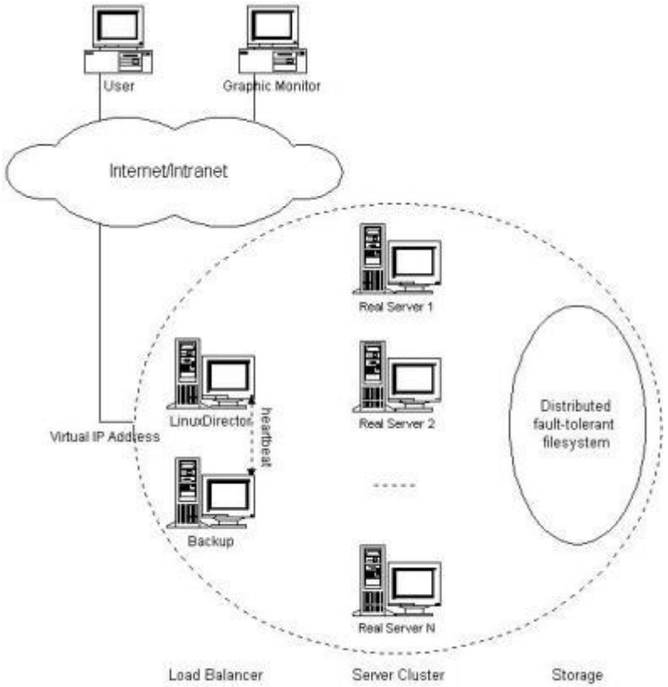
Linux Virtual Server



What is LVS?

LVS (Linux Virtual Server) implements transport-layer load balancing inside the Linux kernel, so called Layer-4 switching.

LVS running on a host acts as a load balancer at the front of a cluster of real servers, it can direct requests for TCP/UDP based services to the real servers, and makes services of the real servers to appear as a virtual service on a single IP address.



LVS has been in active use for 14 years

"Yesterday at DockerCon Europe, Andrey Sibiryov, a senior engineer at ***Uber*** *Technologies, demonstrated how to improve load-balancing performance using an open-source technology that's been part of the Linux kernel for more than a decade — IPVS."*

*"****Wikimedia*** *uses LVS for balancing traffic over multiple servers, see also load balancing architecture"*



NPS-400: a Network Processor

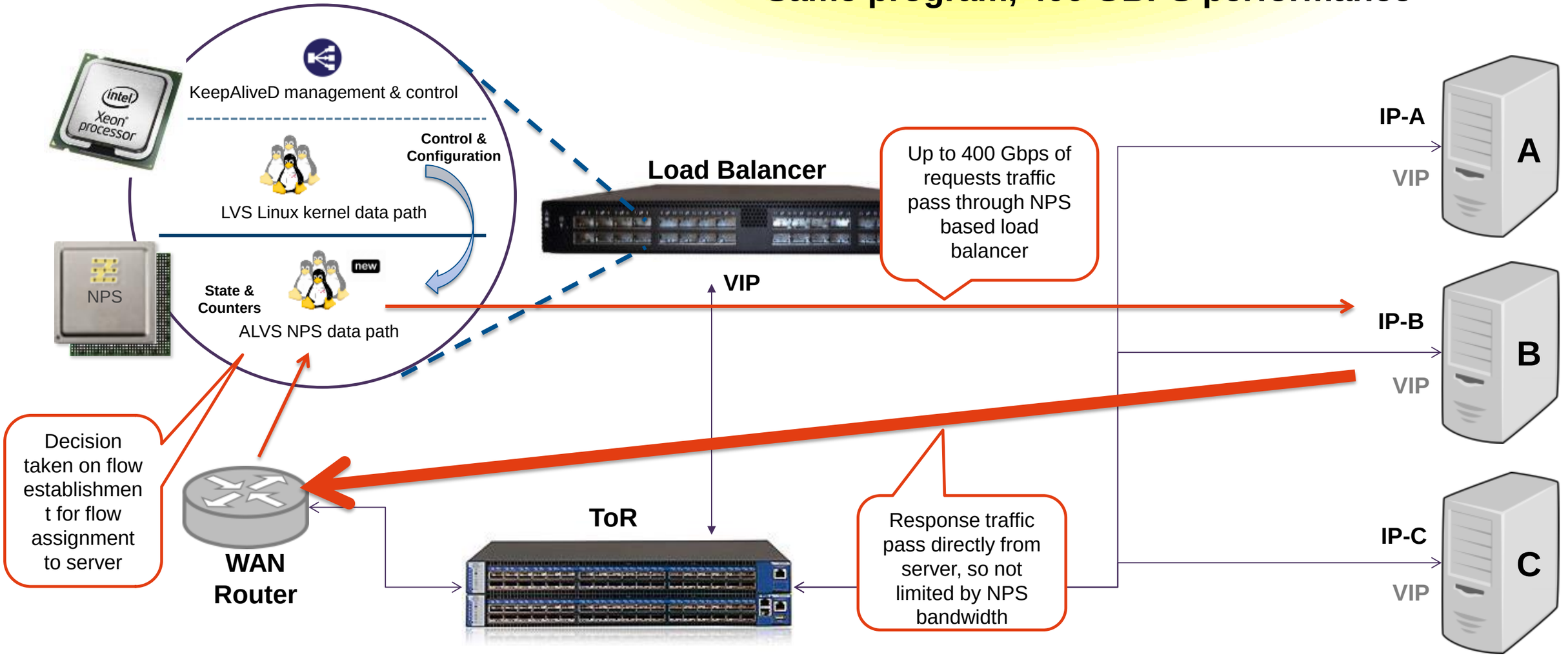
- NPS-400 is a Network Processor
  - Think “GPU, but for networking”
  - An NPU let’s you program your network by writing a program that processes packets at data center line rates
- NPUs used to be part of the secret sauce of carrier equipment
  - e.g. NP-5, NPS predecessor, is part of the Cisco ASR-9K service router shown here
  - These programmable devices were “buried” inside proprietary silos
- We are bringing them into the open and into the data center
  - White box systems from MLNX and ODM
  - OpenNPU – Open Source (GPL v2/BSD) SDK



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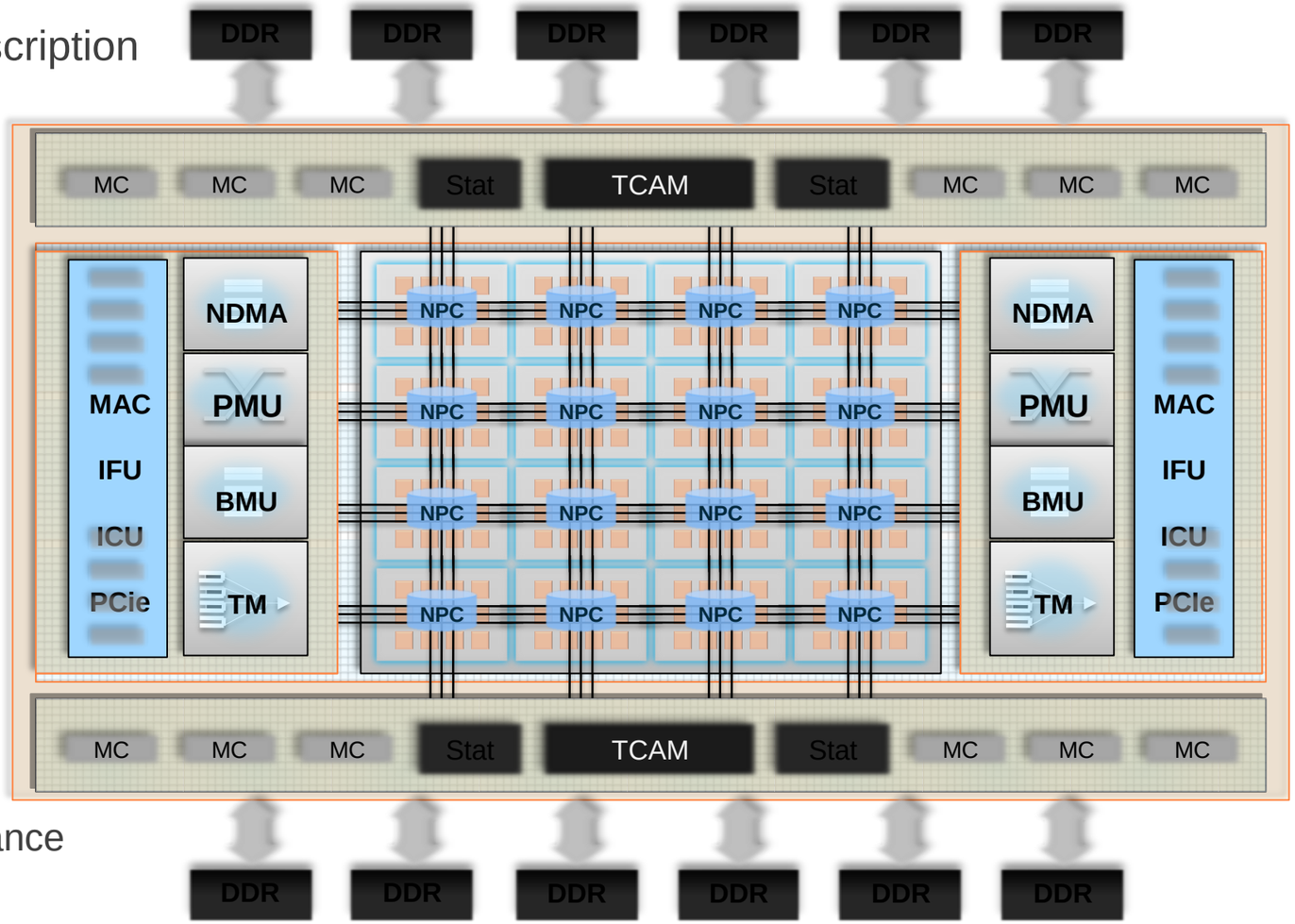
Accelerated Linux Virtual Server

ALVS is LVS with the data path running on a network processor.
Same program, 400 GBPS performance

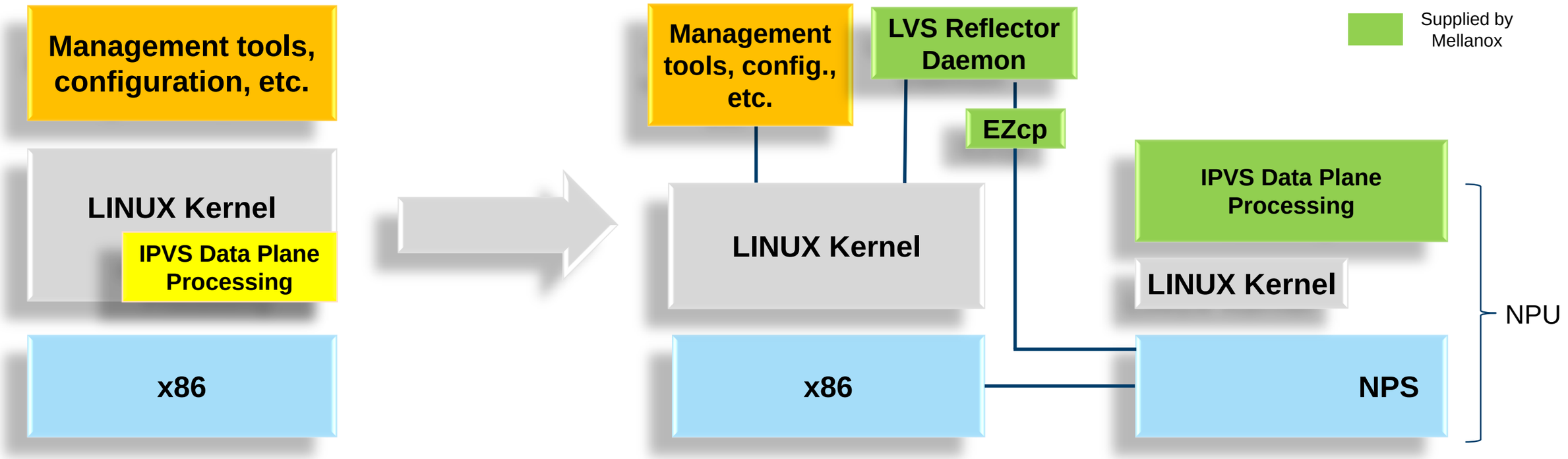


NPS-400 Main Features

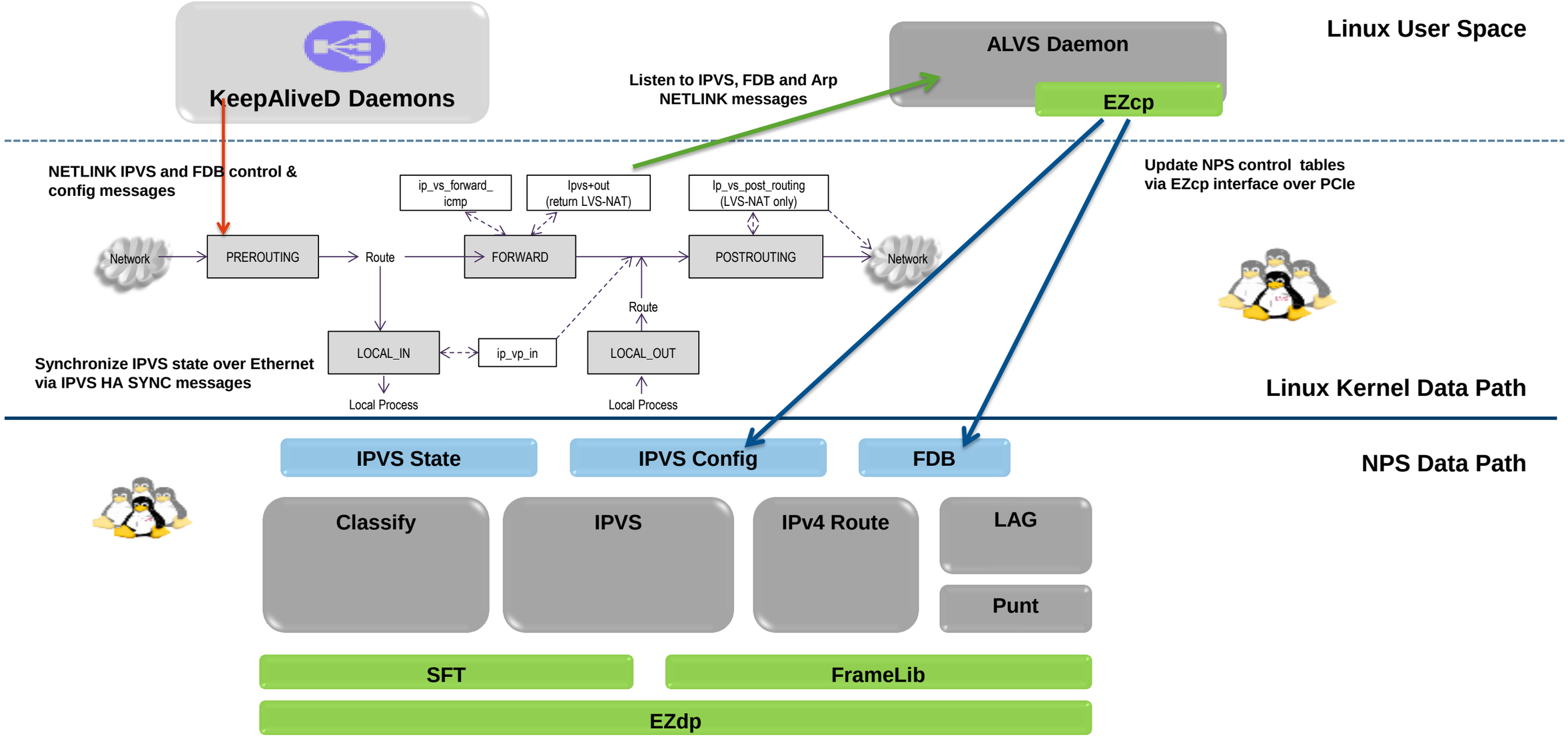
- 400 Gbps line rate
  - 600Mpps wire speed with up to 960 Gbps oversubscription
- Hardware Traffic Manager
  - 1M queues, 5-level H-QoS
- 960Gbps of network I/O
  - Including10GE, 40GE, 100GE, 400G
- 256 CTOP cores – 4,096 CPU (SMT threads)
  - Specialized instruction set for network processing
  - Runs SMP Linux (we're upstream)
- Hardware acceleration engines
  - Crypto (180 Gbps of IPsec), buffer allocations
  - Network order engines, DPI, TCAM
- Commodity DDR (96 GB)
  - Unlimited tables, states, counters at wire-speed performance
- C on Linux programmable
  - Not an ASIC controlled by Linux, it is a processor that runs Linux



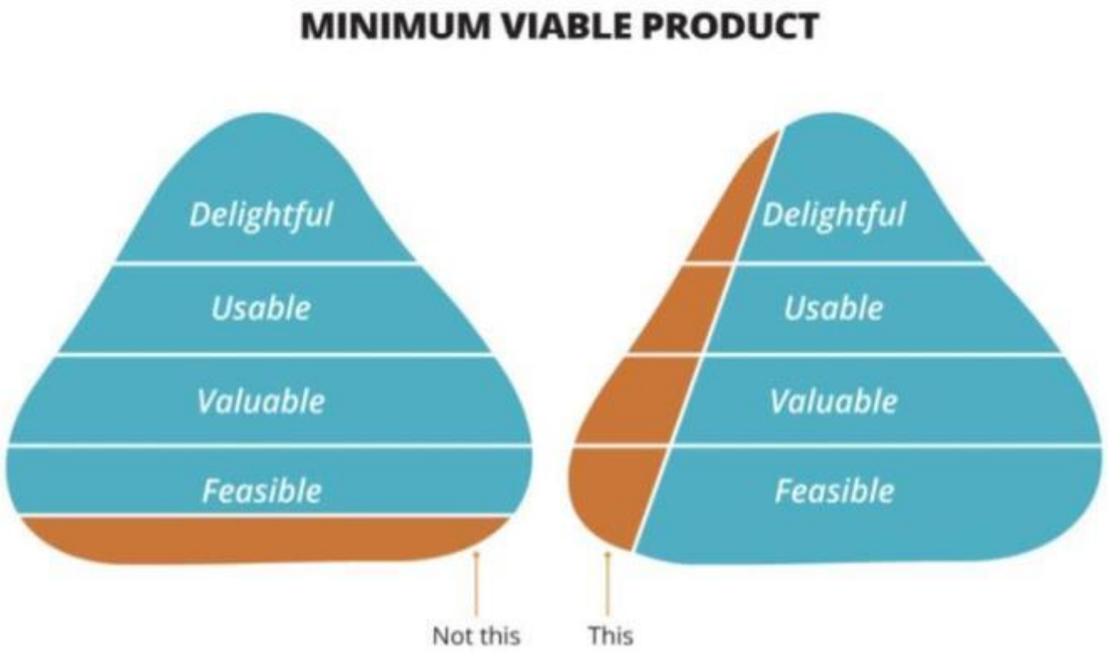
LVS to ALVS Software Migration



Detailed Software Architecture



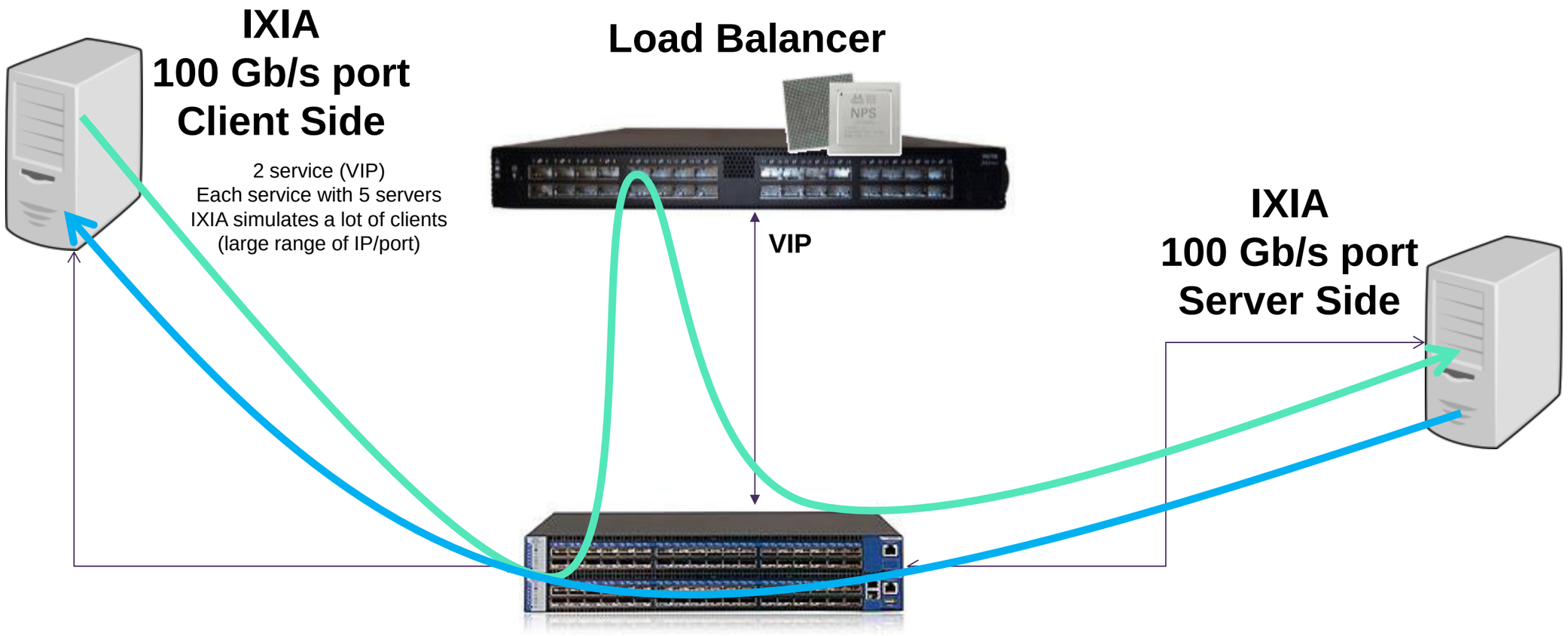
Minimal Viable Product



- **Minimal**
  - Single forwarding mode out of 3
  - Three scheduling algorithms out of 10
  - TCP/IPv4 (will add SCTP, UDP and IPv6 later)
- **Viable**
  - LVS look and feel
    - Same API, same CLI , same log mechanism
    - Integrates with unmodified management plane
  - Robust
    - Resilient. Cover the corner cases
      - Testing already revealed one bug in LVS itself…
    - Supports passive/active fail over
  - **Product**
    - x400 Performance
    - Scales with your ToR switch

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ALVS Test Setup



Test limited by testing equipment scale

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Performance



Criteria	Lab test 25% capacity	Simulation 100% capacity
Concurrent connections	30 M	128 M (200 M)
Connection setup rate	1 M/s	3 M/s
Requests bandwidth	75 Gbps	400 Gbps

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Connecting NPUs to Linux networking stack

- It’s a useful thing to do
  - If you need an L4 load balancer and love LVS, running it at 400 Gbps / 200 M connection on an open source platform is useful
- We need to put the low level NPU driver into the kernel
  - Since NPU is a programmable entity possibly remoteproc subsystem is the right way
- We need to figure out how to hook NPU into network stack
  - Switchdev? XDP? Something else?

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Vision Architecture

VNFs
Layers 2 - 7

OPNFV g-API

Remote



User

NOS
Layers 2 - 7

Open Network Services Interface API



Kernel

OpenNPU API
"The CUDA of NPUs"

Open NPU Control API

Linux net stack
Layers 2 - 7

NPS driver

switchdev



NPS

Mellanox provided Middleware

Stateful connection tracking, DPI Application Recognition, Crypto

Commercial third party data plane

Custom data plane



Open NPU Data Plane API

Linux

Connecting NPUs to Linux networking stack – cont.

- The ALVS data path program ended up very different than IPVS code
  - The architecture of an NPU is very different than a CPU + NIC
    - HW engines for packet scheduling, order restoration, memory architecture that does not rely on caches
    - Program ended up very similar in design to Google Maglev, with HW engines taking place of some of the code blocks
  - This has implications on ideas such as using eBPF/XDP to bring NPU into kernel
    - Yes, you can run the eBPF bytecode, but the program is written under different assumption
- We ran into networking stack scaling issues when trying to synchronize state with the NPU
  - IPVS slowed down to a crawl way before we reached 30 M flows
  - What does it mean when NPU slave device can hold more state than the OS on the host?

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Thank you!



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OPENNPU http://www.opennpu.org

ALVS: https://github.com/Mellanox/ALVS